

Admin and Introduction to this Module

Natural Language Processing, Week 1, Summer 2026

Dr. Paul Nulty

Lecturer

School of Computing and Mathematical Sciences

Birkbeck, University of London

Welcome to Natural Language Processing (NLP)

- Live sessions on Fridays in the summer term (term 3) in 2026, from 6pm to 9pm.

Welcome to Natural Language Processing (NLP)

- Live sessions on Fridays in the summer term (term 3) in 2026, from 6pm to 9pm.
- The room is Malet St /405

Welcome to Natural Language Processing (NLP)

- Live sessions on Fridays in the summer term (term 3) in 2026, from 6pm to 9pm.
- The room is Malet St /405
- Live sessions streamed online and are recorded.

Welcome to Natural Language Processing (NLP)

- Live sessions on Fridays in the summer term (term 3) in 2026, from 6pm to 9pm.
- The room is Malet St /405
- Live sessions streamed online and are recorded.

Modes of delivery in live sessions:

- Lecture parts: new information guided by slides (approx. 50 minutes)
- Lecture continues, plus 'seminar' and demonstration parts: coding examples and discussion (lead by the Lecturer, approx. 50 minutes).

Modes of delivery in live sessions:

- Lecture parts: new information guided by slides (approx. 50 minutes)
- Lecture continues, plus 'seminar' and demonstration parts: coding examples and discussion (lead by the Lecturer, approx. 50 minutes).
- Coding practice in lab (practical work in Python, with help by the lecturer and TA approx. from 8pm).

Lab sessions are not assessed (you do not need to hand in or present your solutions), but working on the exercises helps to understand the taught concepts (and is excellent coursework preparation).

About us

- Lecturer/module convener:
Paul Nulty
`p.nulty@bbk.ac.uk`
- <https://www.bbk.ac.uk/our-staff/profile/9304650/paul-nulty>
- Teaching Assistant for 2025/26:
Harsh Jain
`h.jain@bbk.ac.uk`

About us

- Lecturer/module convener:
Paul Nulty
`p.nulty@bbk.ac.uk`
- <https://www.bbk.ac.uk/our-staff/profile/9304650/paul-nulty>
- Teaching Assistant for 2025/26:
Harsh Jain
`h.jain@bbk.ac.uk`

Topics we will cover

- Natural language processing is a subfield of computer science (not linguistics)

Topics we will cover

- Natural language processing is a subfield of computer science (not linguistics)
- Most modern NLP research studies machine learning methods for processing natural language

Topics we will cover

- Natural language processing is a subfield of computer science (not linguistics)
- Most modern NLP research studies machine learning methods for processing natural language
- However

Topics we will cover

- A good understanding and knowledge of the properties of language and the engineering challenges specific to language is very important both for commercial applications and developing innovative methods.

Topics we will cover

- A good understanding and knowledge of the properties of language and the engineering challenges specific to language is very important both for commercial applications and developing innovative methods.
- My own research focus is on using computers to understand language, and applications for NLP in humanities and social science, not developing machine learning methods for NLP

Topics we will cover

- A good understanding and knowledge of the properties of language and the engineering challenges specific to language is very important both for commercial applications and developing innovative methods.
- My own research focus is on using computers to understand language, and applications for NLP in humanities and social science, not developing machine learning methods for NLP
- You have other modules that focus on formal computer science and machine learning

Topics we will cover

- A good understanding and knowledge of the properties of language and the engineering challenges specific to language is very important both for commercial applications and developing innovative methods.
- My own research focus is on using computers to understand language, and applications for NLP in humanities and social science, not developing machine learning methods for NLP
- You have other modules that focus on formal computer science and machine learning

Topics we will cover

- Therefore, we will focus on methods in computer science that help us to understand language, and to use language to understand the world.

Topics we will cover

- Therefore, we will focus on methods in computer science that help us to understand language, and to use language to understand the world.
- However, you will of course learn about large language models and transformers and how to apply them.

Topics we will cover

- Therefore, we will focus on methods in computer science that help us to understand language, and to use language to understand the world.
- However, you will of course learn about large language models and transformers and how to apply them.

Topics Week 1: Introductions

- Introduction to module

Topics Week 1: Introductions

- Introduction to module
- Overview of AI and NLP

Topics Week 1: Introductions

- Introduction to module
- Overview of AI and NLP
- Introduction to linguistics

- Lexical semantics

Topics Week 2: Words

- Lexical semantics
- Wordnet

- Lexical semantics
- Wordnet
- Distributional semantics

Topics Week 2: Words

- Lexical semantics
- Wordnet
- Distributional semantics
- Word vectors (word embeddings)

- Syntax and parsing, argument structure, stylometry

- Syntax and parsing, argument structure, stylometry

- Document classification and clustering

- Document classification and clustering
- Sentiment analysis

- Document classification and clustering
- Sentiment analysis

- Topic modeling

- Topic modeling

- Document and paragraph embeddings, BERT

- Document and paragraph embeddings, BERT
- Transformers

Topics Week 6: Deep Neural Networks

- Document and paragraph embeddings, BERT
- Transformers
- encoder/decoder models

Topics Week 7: Information retrieval

- indexing and retrieval

Topics Week 7: Information retrieval

- indexing and retrieval
- search engines

Topics Week 7: Information retrieval

- indexing and retrieval
- search engines
- vector databases
- Retrieval Augmented Generation

Topics Week 8: Large Language Models

- decoder only models

Topics Week 8: Large Language Models

- decoder only models
- zero-shot learning and prompting

Topics Week 9: Recent Advances

- Zero-shot learning

Topics Week 9: Recent Advances

- Zero-shot learning
- post-training

Topics Week 9: Recent Advances

- Zero-shot learning
- post-training
- alignment

Topics Week 10: Applications and Philosophical background

- Applications of NLP in humanities and social sciences

- Programming portfolio (60%)
- in-class test (40%)

- Programming portfolio (60%)

- Programming portfolio (60%)
- Released week 6, due week 9 (exact dates tbd)

- Programming portfolio (60%)
- Released week 6, due week 9 (exact dates tbd)
- A series of practical exercises to be completed in Python

- Programming portfolio (60%)
- Released week 6, due week 9 (exact dates tbd)
- A series of practical exercises to be completed in Python
- Based on material from weeks 1-6

- Programming portfolio (60%)
- Released week 6, due week 9 (exact dates tbd)
- A series of practical exercises to be completed in Python
- Based on material from weeks 1-6

- Moodle is the primary mode of communication outside of class time

- Moodle is the primary mode of communication outside of class time
- Please ensure that you receive announcements from Moodle (or check it regularly)

- Moodle is the primary mode of communication outside of class time
- Please ensure that you receive announcements from Moodle (or check it regularly)
- If you have a general question, consider posting it on the Moodle forum

- Moodle is the primary mode of communication outside of class time
- Please ensure that you receive announcements from Moodle (or check it regularly)
- If you have a general question, consider posting it on the Moodle forum
- Moodle tiles with pre and post session activities will appear each week

- In class, the best time to talk to us is during the lab session

- In class, the best time to talk to us is during the lab session
- If you have a question outside class that you don't want to post on the forum:

- In class, the best time to talk to us is during the lab session
- If you have a question outside class that you don't want to post on the forum:
 - p.nulty@bbk.ac.uk
 - h.jain@bbk.ac.uk

- Attending remotely is convenient for you, and you will get all the information you need for the assessment
- It is difficult to give technical or programming help remotely

- Attending remotely is convenient for you, and you will get all the information you need for the assessment
- It is difficult to give technical or programming help remotely
- It is difficult to align in-person and online discussion

- Attending remotely is convenient for you, and you will get all the information you need for the assessment
- It is difficult to give technical or programming help remotely
- It is difficult to align in-person and online discussion
- But you can participate remotely in discussions with us and each other on Teams

- Attending remotely is convenient for you, and you will get all the information you need for the assessment
- It is difficult to give technical or programming help remotely
- It is difficult to align in-person and online discussion
- But you can participate remotely in discussions with us and each other on Teams